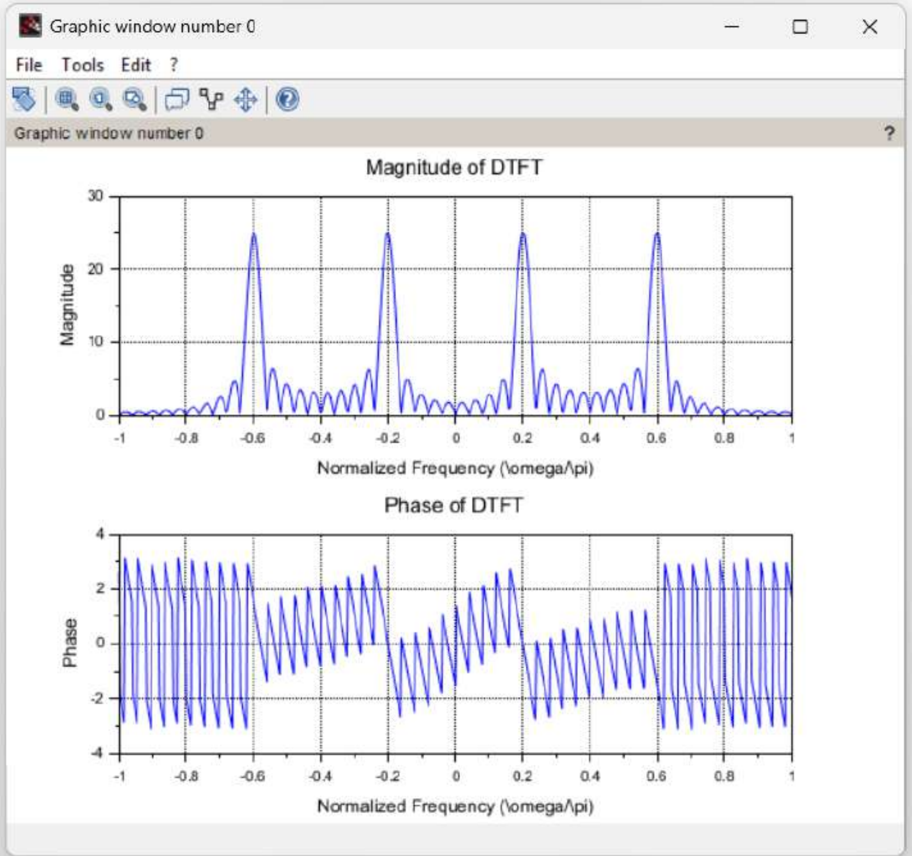
EXP_8.2 SS.sci (E:\EXP_8.2 SS.sci) - SciNotes

EXP_8.2 SS.sci

```
clc;
clear;
clf;

// Parameters
N = 50;
f0 = 0.1;
f1 = 0.3;

n = 0:N-1;

// Signal
x = cos(2*%pi*f0*n) + sin(2*%pi*f1*n);

// Frequency vector
omega = -%pi:0.01:%pi;

X = zeros(omega);

for k = 1:length(omega)
    X(k) = sum(x .* exp(-%i * omega(k) * n));
end

// Magnitude
subplot(2,1,1);
plot(omega/%pi, abs(X));
xlabel("Normalized Frequency (\omega/\pi)");
ylabel("Magnitude");
title("Magnitude of DTFT");
xgrid();

// Phase
subplot(2,1,2);
plot(omega/%pi, atan(imag(X), real(X)));
xlabel("Normalized Frequency (\omega/\pi)");
ylabel("Phase");
title("Phase of DTFT");
xgrid();
```



exp 8.3.sci (C:\Users\imjan\OneDrive\Documents\exp 8.3.sci) - SciNotes

```
// Define parameters
N = 31; // Total number of samples
n = 0:N-1; // Sample index
n1 = 10; // Start of rectangular pulse
n2 = 20; // End of rectangular pulse

// Define the rectangular pulse x[n]
x = zeros(1, N); // Initialize the signal with zeros
x(n1+1:n2+1) = 1; // Set values between n1 and n2 to 1

// Compute DTFT
omega = linspace(-%pi, %pi, 1000); // Frequency range
X = x * exp(-%i * n' * omega); // DTFT calculation

// Plot the magnitude of DTFT

figure();
plot(omega/%pi, abs(X));
xlabel('Normalized Frequency (\omega/\pi)');
ylabel('|X(e^{j\omega})|');
title('Magnitude of DTFT of Rectangular Pulse');

// Plot the phase of DTFT
figure();
plot(omega/%pi, atan(imag(X)./real(X)));
xlabel('Normalized Frequency (\omega/\pi)');
ylabel('Phase of X(e^{j\omega})');
title('Phase of DTFT of Rectangular Pulse');

// Analyze the results in the Scilab console or report
```